

PHYSICAL CONNECTION TEST

1. Types of medium:-

1. Light signal $\rightarrow$ Fiber optical cable [FOC]
2. Electric signal $\rightarrow$ Copper cable [CWP]
3. Radio frequency $\Rightarrow$ Wireless

2. Write down Switch 8-pins

1. + R

2. - R

3. + T

4. + V

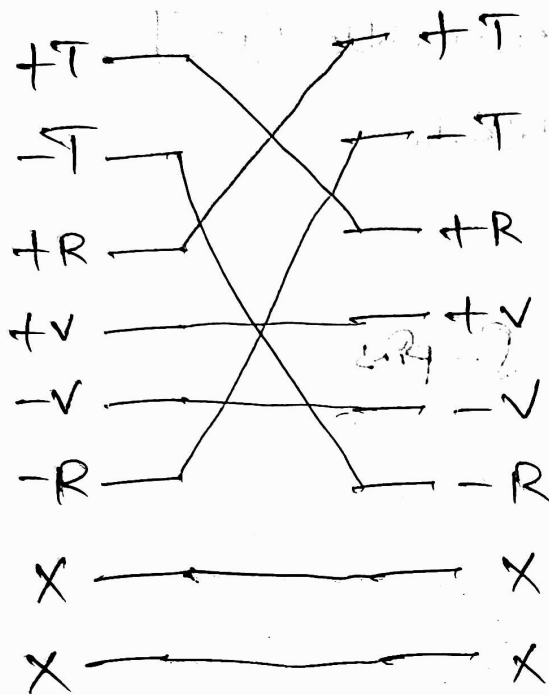
5. - V

6. - T

7. X $\rightarrow$ Empty

8. X $\rightarrow$ Empty

3. Connecting UTP Cable between two devices:-



4. Explaining Auto MDIX & Automatic-Medium Dependent Interface Crossover

It is a feature in ethernet ports that automatically detects whether the connected device needs a straight through or crossover cable.

With this feature the ports automatically adjust the transmit/receive pairs, so either straight through or crossover cable can work.

5. POE Injector-

It is a network device that combines electrical power and network data into a single ethernet cable to provide both power and connectivity to a POE-enabled device.

6. POE Switch:-

This type of switch provides both electric ^{power} and network data through a single ethernet cable to a POE-enabled device.

7. Types of NICs & Functions :-

NIC $\Rightarrow$ Network Interface Card

Types:-

- * Ethernet NIC
- * Wireless NIC
- * Fibre optic NIC
- * USB NIC
- * Virtual NIC

Functions:-

- * Network Connectivity
- * Data transmission & Reception
- * MAC Address - It has unique address
- * Data Conversion
- * Frame handling
- * Error Detection
- * Speed/duplex negotiation